Team Project Combined Report

Insect Pest Classification Using Deep Learning

CSE 429: Computer Vision and Pattern Recognition

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Team Members

#	Name	Student ID	Focus / Model
1	Hossam Eldin	120220313	RAN on D0 Dataset
2	Ziad Reda Mohamed	120220348	Project Overview & Dataset Preparation
3	Mohamed Tareq Farouq	120220307	Deep PCSA + RAN Hybrid on IP102
4	Youssef Ibrahim	120220298	ResNet50 + FPN on IP102
5	Mohamed Ahmed Abdalfattah	120220328	MMAL-Net Development & Optimization
6	Abdelrahman Saeed	120220303	Multi-Model Comparison & Deployment

Section 1 — Hossam Eldin (ID: 120220313)

Topic: Insect Pest Classification Using Residual Attention Network (RAN) on the D0 Dataset

Abstract

This section presents Hossam Eldin's implementation of a Residual Attention Network (RAN) for automated insect pest classification on the D0 benchmark dataset. The model combines residual learning with soft-attention mechanisms to focus on discriminative insect regions. Trained for 5 epochs, the model achieves a final validation accuracy of 69.95% and a test accuracy of 69.00%, demonstrating strong learning progression from 15.27% in epoch 1.

Dataset

The D0 dataset is a publicly available benchmark for insect pest recognition containing 4,508 images belonging to 40 insect pest species captured in natural field environments. The dataset was split following a 70:21:9 ratio (train:test:valid).

Split	Images	Classes	Batches (BS=56)
Train	≈3,156	40	57
Validation	≈316	40	–
Test	≈1,036	40	–
Total	4,508	40	–

Data Preprocessing & Augmentation

- Images resized to 256×256; validation/test resized to 224×224 (no augmentation).
- RandomResizedCrop (224×224, scale [0.75, 1.0]), random horizontal flip (p=0.5), rotation (±15°).
- ColorJitter (brightness, contrast, saturation, hue), random affine translation/scaling.
- RandomErasing (p=0.25) to improve attention robustness.
- ImageNet normalization: mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225].

Model Architecture

The Residual Attention Network (RAN), originally proposed by Wang et al. (2017), augments a standard ResNet-50 backbone with attention modules that learn soft spatial masks. Each module has two parallel branches: a trunk branch (two stacked residual blocks) and a mask branch (lightweight convolutional path with Sigmoid activation). The module output is: $H(x) = (1 + M(x)) \times F(x)$, where the additive 1 ensures trunk features are preserved even when the mask is zero.

Training Configuration

Parameter	Value
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-4}
Batch Size	56
Epochs	5

LR Scheduler	ReduceLROnPlateau (patience=2, factor=0.5)
Loss Function	Cross-Entropy
Dropout	0.5

Results

Epoch	Train Loss	Train Acc (%)	Val Acc (%)	Best?
1	3.5368	13.38	15.27	
2	2.4358	33.41	39.16	
3	1.8548	49.22	53.20	
4	1.2988	61.68	60.34	
5	1.0509	69.00	69.95	✓

The model improved consistently at every epoch with no signs of overfitting. Notably, validation accuracy slightly exceeded training accuracy in the final epoch (69.95% vs. 69.00%), suggesting good generalisation. Best Validation Accuracy: 69.95% | Test Accuracy: 69.00%.

Conclusion

The RAN model leverages soft-attention masks inserted after each ResNet-50 stage to guide feature learning toward discriminative insect regions. Over 5 epochs, training loss dropped from 3.54 to 1.05 and validation accuracy rose from 15.27% to 69.95%, demonstrating rapid convergence. Future work will extend training duration, explore heavier augmentation, and incorporate the ensemble technique from the reference paper.

Section 2 — Ziad Reda Mohamed (ID: 120220348)

Topic: Project Overview — Insect Pest Classification Using CNN Models

Introduction

Insect pests are a major threat to agricultural production, causing significant plant damage and productivity loss. Manual insect recognition is difficult, time-consuming, and expensive. This section, prepared by Ziad Reda Mohamed, provides a broad overview of the project, covering dataset preparation, model study, and ensemble learning analysis based on the paper 'An Efficient Insect Pest Classification Using Multiple CNN-Based Models'.

Project Objectives

- Prepare and organise the IP102 dataset correctly.
- Prepare the D0 dataset directly for training.
- Apply data preprocessing and augmentation techniques.
- Study CNN-based architectures for insect pest classification.
- Analyse the performance of different deep learning models.
- Understand ensemble learning techniques for improving accuracy.

Datasets

IP102 Dataset:

A large-scale benchmark with over 75,000 images across 102 insect classes. Challenges include complex backgrounds, small objects, class imbalance, and high inter-species similarity. Split: 45,095 training / 7,508 validation / 22,619 testing images.

D0 Dataset:

Contains 4,508 images across 40 insect pest species captured in natural environments, split 7:1:2 (train:validation:test).

Deep Learning Models Studied

Three primary architectures were analysed: (1) Residual Attention Network (RAN) — uses attention mechanisms to focus on important insect regions and suppress noisy backgrounds; (2) Feature Pyramid Network (FPN) — handles insects at different scales by building multi-level feature pyramids; and (3) Multi-branch Multi-scale Attention Learning Network (MMAL-Net) — designed for fine-grained classification using multiple branches and attention mechanisms.

Experimental Results

Dataset	Metric	Best Result
IP102	Accuracy	74.13%
IP102	MF1 Score	67.65%
D0	Accuracy	99.78%
D0	MF1 Score	99.68%

The ensemble model combining RAN, FPN, MMAL-Net, and ResNet-50 achieved the highest accuracy, significantly outperforming any single model and prior methods.

Conclusion

The project demonstrated that combining multiple CNN models via ensemble learning provides superior performance for insect pest classification. The ensemble approach consistently outperformed single-model approaches on both datasets, validating the effectiveness of multi-model fusion strategies.

Section 3 — Mohamed Tareq Farouq (ID: 120220307)

Topic: Residual Attention Network for Agricultural Pest Classification on IP102

Abstract

Mohamed Tareq Farouq details the architectural progression of a deep learning model trained on the highly imbalanced IP102 dataset. The research traces the evolution from a baseline model trained from scratch, to a standard ResNet-50 backbone, and finally to a novel Deep PCSA + RAN Hybrid architecture, reaching a peak validation accuracy of 68.42% before encountering class-imbalance overfitting.

Phase 1: Baseline Training Without a Backbone

Standard CNNs were trained from scratch without pretrained ImageNet weights. Without prior feature representations, the network struggled to differentiate complex insect features from textured backgrounds. The model suffered catastrophic failure on minority classes and plateaued early, despite low computational cost.

Phase 2: Transfer Learning with ResNet-50

A ResNet-50 architecture with IMAGENET1K_V1 weights was introduced, with the final fully-connected layer replaced to output 102 classes. While validation accuracy improved significantly, the model suffered from 'attention dilution' — it extracted features from leaves, background dirt, or water droplets rather than isolating the pest itself.

Phase 3: Deep PCSA + RAN Hybrid

To solve attention dilution, a Parameter-Free Channel Spatial Attention (PCSA) module was embedded inside every bottleneck block of the ResNet-50 trunk, dynamically filtering background noise at every feature extraction stage. A True Residual Attention Network (RAN) module was then attached to the 2048-channel output for global spatial refinement before classification.

Training Challenges & Hardware:

- Combining pretrained ImageNet weights with randomly-initialised PCSA/RAN weights caused training instability.
- Resolved via Differential Learning Rates: ResNet trunk frozen at 1e-5; new attention modules at 1e-4.
- Hardware: Kaggle Tesla T4 GPU (15GB VRAM); ~11 min 40 sec per epoch.

Results & Overfitting Wall

Peak validation accuracy of 68.42% was achieved at Epoch 7. By that point training accuracy reached 89.20% while validation loss spiked to 1.4182, indicating the model memorised dominant majority-class pixel configurations and began misclassifying minority classes.

Future Work

- Hardware Optimisation: Use nn.DataParallel across both Kaggle T4 GPUs.

- Mathematical Balancing: WeightedRandomSampler with square-root smoothed inverse frequencies.
- Robust Augmentation: Aggressive spatial augmentations for balanced minority classes.

Section 4 — Youssef Ibrahim (ID: 120220298)

Topic: ResNet50 with Feature Pyramid Network (FPN) on IP102

Introduction

Youssef Ibrahim's contribution focuses on applying a ResNet50 model combined with a Feature Pyramid Network (FPN) for image classification on the 102-class IP102 dataset. FPN improves feature extraction by combining low-level and high-level feature maps from different network layers, enabling the model to capture both fine detail and semantic information simultaneously.

Training Setup

The model was trained for 10 epochs with a pretrained ResNet50 base. The fully-connected layer was replaced to match the 102 output classes. Both training loss and accuracy were monitored throughout.

Training Results

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	2.0864	0.4734	1.6838	0.5599
2	1.5305	0.5828	1.5664	0.5802
3	1.3221	0.6306	1.5084	0.6107
4	1.1923	0.6604	1.4231	0.6265
5	1.0841	0.6849	1.3847	0.6383
6	0.9915	0.7086	1.4858	0.6371
7	0.9234	0.7280	1.3643	0.6496
8	0.8449	0.7471	1.3888	0.6589
9	0.7871	0.7582	1.4151	0.6584
10	0.7292	0.7752	1.4613	0.6540

Comparison with Paper Results

Metric	Our Model	Paper Target
Best Validation Accuracy	65.89%	70.42%

The ResNet50 + FPN model showed continuous improvement during training, achieving a best validation accuracy of 65.89% across 102 classes — close to the paper's reported 70.42%. FPN's multi-scale feature extraction contributed meaningfully to the final classification performance.

Conclusion

Training accuracy increased steadily while validation accuracy plateaued at 65.89%. The use of FPN helped the model extract multi-scale features, improving overall classification performance. Further hyperparameter tuning and extended training are expected to close the gap to the paper baseline.

Section 5 — Mohamed Ahmed Abdalfattah (ID: 120220328)

Topic: MMAL-Net Development, Optimisation, and Training on IP102

Individual Contribution Statement

Mohamed Ahmed Abdalfattah's primary responsibility was the end-to-end development, debugging, and training of the Multi-branch Multi-scale Attention Learning (MMAL-Net) model — the 'fine-grained expert' of the classification system, designed to distinguish between highly similar insect species in IP102.

Architecture Integration

The multi-branch architecture was implemented with three specialised branches: a Global Branch for full-image context, an Object Branch for localised regions, and a Part Branch for fine-grained morphological features. This enables the model to act as a specialist in identifying subtle differences between insect species.

Architectural Optimisation: Feature-Level Cropping

The standard MMAL-Net requires three separate forward passes through the ResNet backbone — computationally expensive. Mohamed Ahmed implemented Feature-Level Cropping:

- Single-Pass Backbone: ResNet-50 runs once on the raw image to generate a global feature map.
- RoIAlign Implementation: torchvision.ops.roi_align extracts Local and Window features from high-level feature maps.
- Efficiency Gain: Reduced convolutional operations by ~60–70%; training time per epoch cut from >3,000s to <1,000s.

Debugging & Environment Compatibility

Significant legacy code issues were resolved, including patching model.py to fix AttributeError bugs caused by deprecated NumPy types (np.int vs int), allowing the code to execute correctly in a PyTorch 2.x environment.

Data Engineering & Hardware Management

The dataset was moved from network storage to local SSD (/kaggle/working/) to eliminate I/O bottlenecks on a Kaggle P100 GPU (16GB VRAM). Input resolution was downscaled to 224×224 while using ROI Align to maintain high-quality localised features.

Experimental Results

Metric	Value	Notes
Peak Raw Accuracy (Global)	64.13%	After 30 epochs
Peak Local Accuracy (Object)	55.94%	Doubled from 25% at start

The convergence of the Local Accuracy curve validates that the attention mechanism successfully learned to localise pests within complex natural backgrounds.

Conclusion

By implementing RoIAlign optimisation, Mohamed Ahmed provided the team with a high-speed, robust classification engine, demonstrating the practical balance between architectural complexity and computational efficiency in real-world computer vision tasks.

Section 6 — Abdelrahman Saeed (ID: 120220303)

Topic: Multi-Model Comparison on IP102 & Streamlit Deployment

Introduction

Abdelrahman Saeed's contribution covers training and comparing multiple pretrained CNN models on the IP102 dataset and subsequently deploying the best-performing models via a Streamlit web application.

Models Evaluated

- ResNet50
- AlexNet
- ResNet50 FPN

All models were implemented using PyTorch.

Training Results

ResNet50 on IP102:

Metric	Value
Final Training Accuracy	80.04%
Best Validation Accuracy	67.99%

AlexNet on IP102:

Metric	Value
Final Training Accuracy	73.70%
Best Validation Accuracy	60.28%

ResNet50 FPN on Paper Dataset (D0):

Metric	Value
Final Training Accuracy	97.86%
Best Validation Accuracy	99.11%

Model Comparison Summary

Model	Dataset	Val Accuracy	Train Accuracy
ResNet50	IP102	67.99%	80.04%
AlexNet	IP102	60.28%	73.70%
ResNet50 FPN	Paper Dataset	99.11%	97.86%

ResNet50 and ResNet50 FPN outperformed AlexNet due to their deeper architectures and stronger feature extraction capabilities through residual connections.

Deployment

After training, the project was deployed using Streamlit. The application was first developed locally using VS Code and later published on GitHub. The app loads the best saved model checkpoints and combines them using ensemble prediction to improve prediction accuracy.

GitHub Repository: https://github.com/Abdelrahmansa04/Insect-Pest-Classification

Conclusion

Across multiple architectures, ResNet50 consistently outperformed the simpler AlexNet on IP102, while ResNet50 FPN achieved near-perfect accuracy on the D0 dataset. Ensemble prediction in the deployed application further enhanced classification reliability, demonstrating an effective end-to-end workflow from model training to production deployment.

Team Results Summary

Student	Model	Dataset	Best Val Acc
Hossam Eldin	RAN	D0	69.95%
Ziad Reda Mohamed	Overview/Ensemble	Both	99.78% (D0)
Mohamed Tareq Farouq	PCSA + RAN Hybrid	IP102	68.42%
Youssef Ibrahim	ResNet50 + FPN	IP102	65.89%
Mohamed Ahmed Abdalfattah	MMAL-Net	IP102	64.13%
Abdelrahman Saeed	Multi-model comp.	IP102/D0	99.11% (D0)